

Patrick Smith

Building production platforms from scratch: APIs, databases, cloud infrastructure, and everything in between

🏠 Hollister, CA
✉️ patrick@nerdgeekhero.com
🐙 github.com/murRAIN

☎️ (415)-660-6195
🌐 linkedin.com/in/patrick-smith-1963608



📄 Summary

I build systems that outlast their origin story. For over 13 years, I served as the sole founding engineer behind a production B2B marketplace, designing and operating the full platform from the ground up. The work was not about flash or novelty, but about reliability, correctness, and systems that continue to function as the business grows and changes.

I owned the full lifecycle, translating loosely defined business needs into durable technical solutions and supporting them in production through real traffic, real money, and real failure modes. I resolved issues wherever they surfaced, from application code and databases to networks and infrastructure. My experience includes Python, PHP, JavaScript, MySQL, AWS and Azure, Stripe-based billing, observability with Grafana and Prometheus, and hands-on security incident response. The proof lives in systems that earned trust by working when it mattered.

🧰 Experience

Founding Engineer

May 2011 – Sept 2024

Home Remodel Buddy, San Francisco, CA

- Independently architected and built B2B marketplace platform for 13 years as sole founding engineer, processing \$2.5M annually across four integrated systems serving 740+ contractors. Owned complete technical stack from requirements gathering through production support, making autonomous decisions on technology selection, system architecture, and implementation approach with minimal supervision.
- Translated business requirements from non-technical stakeholders (executives, sales, operations) into technical specifications and working software. Conducted stakeholder interviews to understand needs, provided feasibility analysis with trade-off recommendations (cost/time/complexity), estimated timelines, and delivered progress updates in business terms focusing on customer impact and revenue implications.
- Independently debugged and resolved complex production incidents spanning multiple system layers: identified $O(n^2)$ MySQL query causing hours-long batch processing using EXPLAIN analysis, diagnosed fail2ban iptables misconfiguration blocking nginx-to-apache communication via systematic performance methodology, and responded to WordPress security compromise by reverse-engineering obfuscated PHP malware—all with 4-hour recovery and zero data loss.
- Managed complete project lifecycle autonomously: gathered requirements from stakeholders, designed technical solutions with architecture diagrams, implemented across backend (PHP/Python APIs), database (MySQL schema), frontend (Knockout.js SPA), and infrastructure (AWS/Azure), deployed with zero-downtime strategies, and provided ongoing support with iterative improvements based on user feedback.
- Built Hercules distribution platform by independently designing sophisticated matching algorithm balancing geographic coverage, product expertise, contractor capacity, 120-day deduplication windows, and value-based prioritization. Explained algorithm trade-offs to business stakeholders using non-technical language, showing how technical decisions prevented duplicate charges while optimizing revenue across 740+ contractors.
- Integrated Stripe payment processing (\$2.5M annually) by independently implementing complete lifecycle: customer creation, charge execution, webhook event handling,

⚙️ Skills

Languages

Python JavaScript
PHP SQL
Bash

Cloud & Infrastructure

AWS Azure
Linux Docker
Git CI/CD

Databases & Data

MySQL Redis
ETL NumPy
Pandas sklearn

APIs & Integration

REST SOAP
OAuth JSON/XML
Webhooks

Monitoring & Security

Grafana Prometheus
IDS PCI
Incident Response

Networking

VPC VPN
DNS Firewalls

👥 References

Christopher
Levine, Director
of Eng

Lyft

415-827-2097,
clevine@keyobjects.com

Barry
Willingham,
CEO

*Funnel
Social*

415-655-1844,
barry@swimmingpoolquotes.com

dispute resolution workflow, and monthly invoicing. Communicated payment system capabilities and limitations to stakeholders, provided financial reconciliation reports, and built self-service contractor portal reducing support burden.

- Deployed infrastructure across AWS and Azure with multi-region availability, then independently evaluated and executed migration to bare metal achieving 60% cost reduction. Presented cost-benefit analysis to executives with ROI calculations, migration timeline, and risk mitigation strategies. Completed migration while maintaining service continuity and stakeholder confidence.
- Responded to production incidents with clear stakeholder communication: provided immediate status updates to affected customers, estimated resolution timelines, explained root causes in business terms, and documented preventive measures. Maintained <15 minute average response time over 13 years of on-call support (nights, weekends, holidays) with 30-minute maximum downtime for critical systems.

Education

A.S Business/Economics

2005-2007

Gavilan College, Gilroy, CA

Computer Science Coursework

2003-2005

California Polytechnic University, San Luis Obispo, San Luis Obispo, CA

CS109x: Introduction to Data Science with Python

July, 2025

HarvardX (Online Learning Initiative of Harvard University), <https://courses.edx.org/certificates/af05411c855d4ab7aaf9c8d90b29bd09>

Machine learning with sklearn, data analysis with Pandas/NumPy, regression and classification models, cross-validation, regularization, hypothesis testing, and statistical modeling

PH125.3x: Data Science: Probability

July, 2025

HarvardX (Online Learning Initiative of Harvard University), <https://courses.edx.org/certificates/73a1329507e642a0977384e4344c650e>

Probability theory, random variables, Monte Carlo simulations, Central Limit Theorem, expected values, standard errors, statistical inference foundations, and sampling models